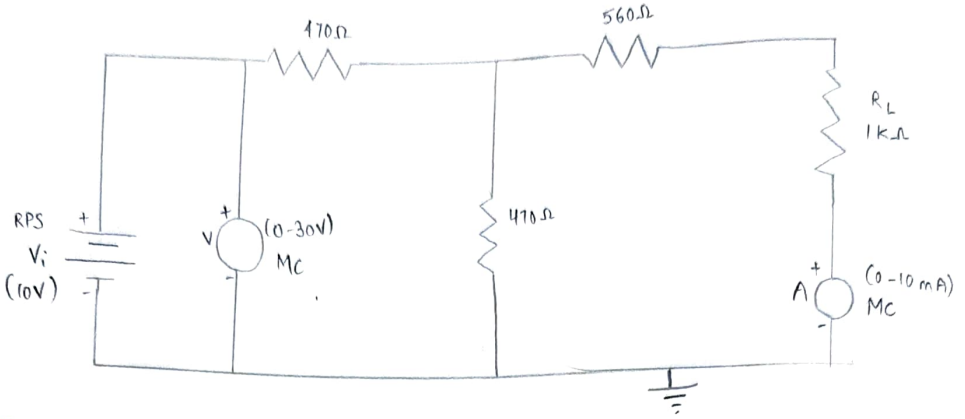
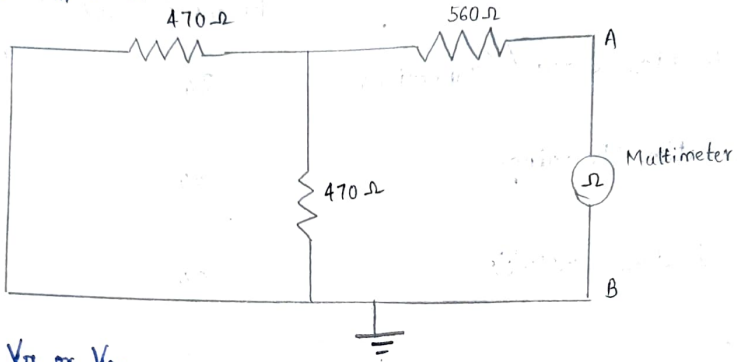


CIRCUIT DIAGRAM:

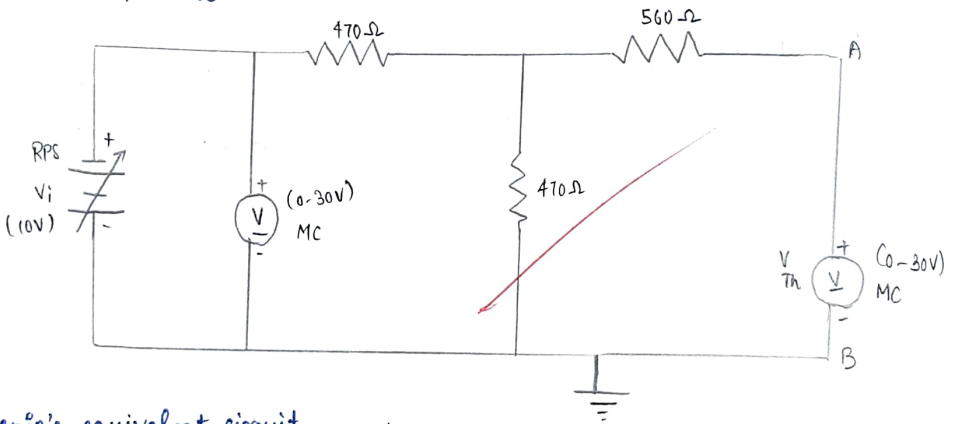
To measure I_L



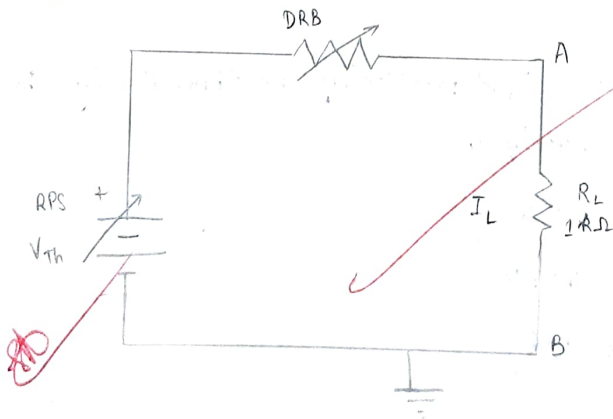
To measure R_{Th} or R_N



To measure V_{Th} or V_{oc}



Thevenin's equivalent circuit



VERIFICATION OF THEVENIN'S THEOREM

AIM:

To verify Thevenin's theorem practically and theoretically for the given DC circuit.

APPARATUS REQUIRED:

SI. NO	APPARATUS	SPECIFICATION	QUANTITY
1.	Regulated Power Supply (RPS)	(0-30V)	1
2.	Voltmeter	(0-30V) MC	1
3.	Ammeter	(0-10mA) MC	1
4.	Resistors	470 Ω , 560 Ω , 1k Ω	2, 1, 1
5.	Bread board	—	1
6.	Multimeter	—	1

PROCEDURE:

1. Make the connections as per the Circuit Diagram: 1.
2. Vary the RPS and set an input voltage of 10V.
3. Note down the voltmeter reading (V_i) and ammeter reading (I_L) in Tabular Column 1.
4. Switch off the supply and make connections for Circuit Diagram 2.
5. Measure the Thevenin's resistance R_{Th} = Norton's Resistance R_N .
6. Switch off the supply and make connections for Circuit Diagram: 3.
7. Set an input voltage of 10V in the RPS and note down the voltmeter readings V_i and V_{Th} ($= V_{oc}$) in Tabular Column 4.
8. Switch off the supply and make connections for Circuit Diagram 4.
9. Set an input voltage of 10V in the RPS and note down the voltmeter reading V_i and Ammeter reading I_N ($= I_{sc}$) in Tabular Column 4.
10. Draw the Thevenin's equivalent circuit and Norton's equivalent circuit as shown in circuit diagrams 5 & 6 respectively.
11. Calculate the I_L value using the formula — Thevenin's Theorem: $I_L = V_{Th} / (R_{Th} + R_L)$, Norton's Theorem: $I_L = I_N * R_N / (R_N + R_L)$
12. Theoretically verify the Norton's theorem.

TABULAR COLUMN: 1

To measure I_L

V_i (volts)	I_L (amps)
10V	2.81 mA

TABULAR COLUMN: 3

To measure V_{Th} or V_{oc}

V_i (volts)	V_{Th} (volts)
10V	5V

TABULAR COLUMN: 2

To measure R_{Th} or R_N

From circuit diagram 2, $R_{Th} = R_N = 795 \Omega$

MODEL CALCULATION:

Practical value of I_L (from tabulation)
 $= 2.3 \text{ mA}$

Verification of Thevenin's theorem

$$I_L = \frac{V_{Th}}{(R_{Th} + R_L)}$$

$$I_L = 2.22 \text{ mA}$$

THEORETICAL CALCULATION of I_L , R_{Th} (R_N)

V_{Th} for the given circuit

By voltage division rule

$$V_{Th} = \frac{V_S \times R_2}{R_1 + R_2}$$

$$V_{Th} = \frac{10V \times 470 \Omega}{(470 + 470) \Omega}$$

$$V_{Th} = 5V$$

(No current flow through R_2 when R_2 is open circuit).

R_{Th} : R_1 & R_2 is parallel when voltage source is short circuited.

$$R_{Th} = \frac{470 \times 470}{470 + 470} + 560 = 235 + 560 = 795 \Omega$$

$$R_{Th} = 795 \Omega$$

Find I_L :

R_{Th} in series with R_L

$$\therefore R_{eq} = R_L + R_{Th} = 1.795 k\Omega$$

$$I_L = \frac{V_{Th}}{R_{eq}} = \frac{5V}{1.795 \times 10^3 \Omega} = 2.79 \text{ mA}$$

$$I_L = 2.79 \text{ mA}$$

SI. NO	Description	Marks Allotted	Marks Obtained
1.	Experimental Setup / Connections	20	20
2.	Observation / Readings	20	20
3.	Calculations / Results	20	20
4.	Viva	20	20
5.	Record	20	20
		Total:	100

100

24/2/26

RESULT:

Thus Thevenin's theorem is verified practically and theoretically.